

Watch-DevKit-HW Fabrication Document

Layer Stack Legend

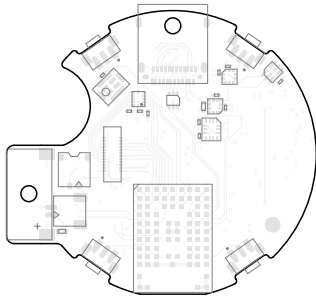
	Material	Layer	Thickness	Dielectric	Type	Gerber
		F.Paste			Paste Mask	
		F.Silkscreen			Legend	GBR
		F.Mask	0,01mm		Solder Mask	GBR
	Copper	F.Cu	0,035mm (1,00oz)		Signal	GBR
	Core		0,196mm	FR4	Dielectric	
	Copper	Inner1	0,035mm (1,00oz)		Plane	GBR
	Prepreg		0,43mm	FR4	Dielectric	
	Copper	Inner2	0,035mm (1,00oz)		Plane	GBR
	Core		0,196mm	FR4	Dielectric	
	Copper	B.Cu	0,035mm (1,00oz)		Signal	GBR
	B.Mask		0,01mm		Solder Mask	GBR
	B.Silkscreen				Legend	GBR
	B.Paste				Paste Mask	

Total thickness: 0,982mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [%]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
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Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- FABRICATE PER IPC-6012A CLASS 2.
- OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

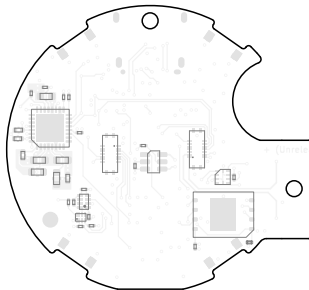
DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- SURFACE FINISH: HAL LEAD-FREE
- SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR WHITE.
- SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING BLACK NON-CONDUCTIVE EPOXY INK.
- ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- PCB MATERIAL REQUIREMENTS:

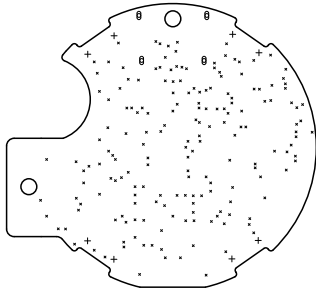
A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
B. Tg 150 C OR EQUIVALENT.
- DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE 40,972 × 37,500 mm
BOARD THICKNESS 0,982 mm
TRACE WIDTH 0,150 mm
TRACE TO TRACE 0,125 mm
MIN. HOLE (PTH) 0,250 mm
MIN. HOLE (NPTH) 0,650 mm
ANNULAR RING 0,100 mm
COPPER TO HOLE 0,125 mm
COPPER TO EDGE 0,125 mm
HOLE TO HOLE 0,250 mm

	Comments: ZSWatch	Company: ZSWatch		Variant: PRELIMINARY	Git Hash: 39449ac
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: ZSWatch-Watch.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 1 of 10

Watch-DevKit-HW Fabrication Document							
Bottom Fabrication (Scale 1:1)							
							
	Comments:	Company:		Variant:	Git Hash:		
	ZSWatch	ZSWatch		PRELIMINARY	39449ac		
	Board Name:	Project Name:					
	Watch-DevKit-HW		ZSWatch Watch-DevKit				
Sheet Title:	File Name:	Designer:	Date:	Revision:			
Bottom Fabrication (Scale 1:1)	ZSWatch-Watch.kicad_pcb	Daniel Kampert	2025-07-13	+ (Unreleased)			
Sheet Path:	Reviewer:		Size:	Sheet:			
			A3	2 of 10			

Drill Drawing L1 - L4 (Scale 1:1)

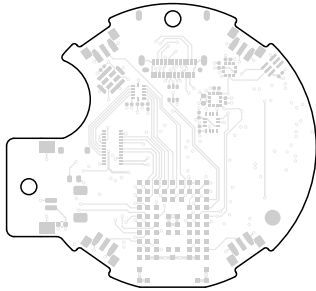


Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	182	0,25mm (9,84mils)	PTH	Round	F,Cu - B,Cu	Via
O	4	0,50mm (19,69mils)	PTH	Slot	F,Cu - B,Cu	Pad
+	8	0,80mm (31,50mils)	PTH	Round	F,Cu - B,Cu	Pad
	Total 194					

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	Sheet Path:		Reviewer:	Size: A3	Sheet: 3 of 10

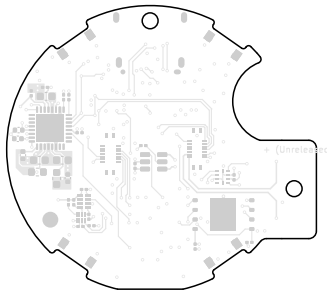
Top Test Points (Scale 1:1)

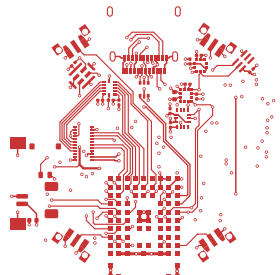


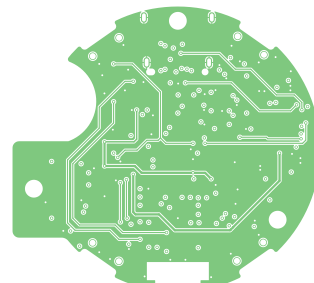
Ref.	Net	X [mm]	Y [mm]
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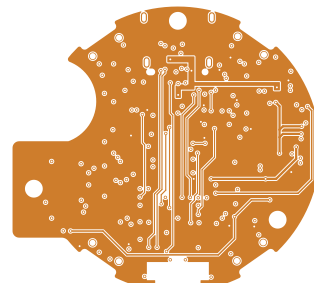
Ref.	Net	X [mm]	Y [mm]
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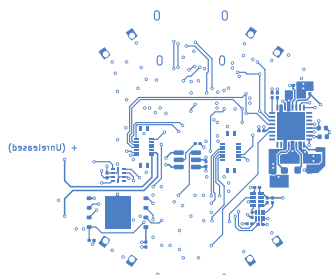
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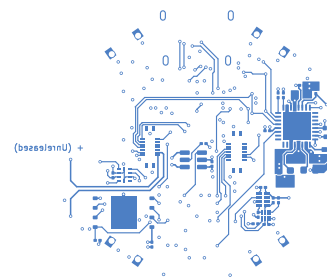
Watch-DevKit-HW Fabrication Document								
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	ZSWatch		ZSWatch		PRELIMINARY		39449ac	
			Board Name:		Project Name:			
			Watch-DevKit-HW		ZSWatch Watch-DevKit			
	Sheet Title:		File Name:		Designer:		Date:	
	Bottom Test Points (Scale 1:1)		ZSWatch-Watch.kicad_pcb		Daniel Kampert		2025-07-13	
	Revision:		Size:		Sheet:			
	+ (Unreleased)		A3		6		of 10	
Sheet Path:		Reviewer:						

Watch-DevKit-HW Fabrication Document										
F.Cu (Scale 1:1)										
										
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			Board Name: Watch-DevKit-HW			Project Name: ZSWatch Watch-DevKit				
	Sheet Title: F.Cu (Scale 1:1)		File Name: ZSWatch-Watch.kicad_pcb		Designer: Daniel Kampert		Date: 2025-07-13		Revision: + (Unreleased)	
	Sheet Path:				Reviewer:		Size: A3		Sheet: 7 of 10	

Watch-DevKit-HW Fabrication Document							
Inner1 (Scale 1:1)							
							

Watch-DevKit-HW Fabrication Document							
Inner2 (Scale 1:1)							
							

A	1	2	3	4	5	6	7	8																			
B																											
C																											
D																											
E																											
F	1	2	3	4	5	6	7	8																			
Watch-DevKit-HW Fabrication Document					B.Cu (Scale 1:1)			<table> <tr> <td data-bbox="1855 1715 2098 1864" rowspan="2"> Comments: ZSWatch </td><td colspan="2" data-bbox="2098 1715 2591 1789"> Company: ZSWatch </td><td data-bbox="2591 1715 2760 1789"> Variant: PRELIMINARY </td><td data-bbox="2760 1715 2890 1789"> Git Hash: 39449ac </td></tr> <tr> <td colspan="2" data-bbox="2098 1789 2591 1864"> Board Name: Watch-DevKit-HW </td><td colspan="2" data-bbox="2591 1789 2890 1864"> Project Name: ZSWatch Watch-DevKit </td></tr> <tr> <td data-bbox="1855 1864 2098 1938"> Sheet Title: B.Cu (Scale 1:1) </td><td data-bbox="2098 1864 2395 1938"> File Name: ZSWatch-Watch.kicad_pcb </td><td data-bbox="2395 1864 2591 1938"> Designer: Daniel Kampert </td><td data-bbox="2591 1864 2703 1938"> Date: 2025-07-13 </td><td data-bbox="2703 1864 2890 1938"> Revision: + (Unreleased) </td></tr> <tr> <td colspan="2" data-bbox="1855 1938 2395 2013"> Sheet Path: </td><td data-bbox="2395 1938 2591 2013"> Reviewer: </td><td data-bbox="2591 1938 2703 2013"> Size: A3 </td><td data-bbox="2703 1938 2890 2013"> Sheet: 10 of 10 </td></tr> </table>	Comments: ZSWatch	Company: ZSWatch		Variant: PRELIMINARY	Git Hash: 39449ac	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit		Sheet Title: B.Cu (Scale 1:1)	File Name: ZSWatch-Watch.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)	Sheet Path:		Reviewer:	Size: A3	Sheet: 10 of 10
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